

	Baseline: equal length, in-register, G-X-Y	(a) Unequal lengths: full-triplet difference	(b) Unequal lengths: incomplete terminal triplet	(c) Wrong/shifted register	(d) Mid-chain periodicity break
Frequency	7 of 80 seqs (8.8%)	19 of 80 seqs (23.8%)	68 of 80 seqs (85.0%)	25 of 80 seqs (31.3%)	5 of 80 seqs (6.3%)
Chain A →					
Chain B →					
Chain C →					
TheBuScr	✓ Builds full structure	✗ Cannot build	✗ Cannot build	✗ Cannot build	✗ Cannot build
	Equal-length, Gly-started, in-register chains satisfy all implicit assumptions; helix type (IR/AR) and stagger are well defined.	Silently truncates all chains to the shortest common triplet-multiple core, discarding valid terminal triplets in chains A and B.	Requires each chain to comprise whole number of triplets; chain B with a trailing partial triplet has no valid X/Y and cannot be parsed.	No frame detection: chunks each chain into Gly-leading triplets regardless of register, producing incorrect backbone geometry for chain C.	Helix type is assigned by propensity-avg over a 5-triplet window; internal periodicity break can result in no valid Gly-led triplet, causing div/0.
CDSM	✓ Builds full structure	✓ Builds full structure	✓ Builds full structure	✓ Builds full structure	✗ Cannot build
	Defaults to native TheBuScr behaviour; no extension needed.	Truncates to common core, then extends geometry type to the dropped triplet and rebuilds natively.	Truncates to common core; extra residues restored via Kabsch-derived extrapolation.	Reframing identifies maximum Gly-starting frame and offsets accordingly, correcting register before build.	A single global reframe cannot resolve a mid-chain break; helix-type assignment still fails. Future work to fix.